Franklin Heng

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CV/ML engineer building solutions across **healthcare**, **climate**, and **applied AI**. Deployed and built products 0 → 1.

WORK EXPERIENCE

Applied ML Engineer, Make The Dot — *September 2025 – January 2026*

- **Owned the GPU inference stack for stable diffusion sketch to render pipeline that produces photorealistic clothing renders in seconds.** The pipeline combines Juggernaut-X v10 SDXL, ControlNet-Union, and custom LoRAs, served through **NVIDIA Triton and TensorRT** (FP16, CUDA graphs, refit) on H100. *[view details]*
- Trained **fabric-specific LoRAs** with Hugging Face Diffusers. Built a **multi LoRA weighted blending** system (for example, faded = light at 0.9 plus bleach at 0.3) that creates new wash variants at inference without retraining.
- Built an automated **A/B image evaluation pipeline** (testing across fabric, clothing, and color) that runs in parallel against the live API in order to evaluate different model performances.

Computer Vision Engineer, Running Tide — *March 2022 – June 2024*

- **Built a computer vision pipeline that verifies ocean carbon removal.** Imagery from custom ocean buoy cameras flows through an AutoML CNN noise filter. Mask R-CNN then segments biomaterials (TensorFlow, OpenCV, scikit-image), and the pixel area over time is fit to an **exponential decay sinking curve,** which feeds into our carbon verification dashboard. Containerized in Docker and orchestrated in Google Cloud Composer and Airflow. *[view abstract]*
- Fine-tuned a **Faster R-CNN** model to detect and count shellfish on a robotic shellfish harvester at **over 90% accuracy and 0.125 s per image** (TensorFlow, OpenCV). Each image contained about 1,000 shellfish, some as small as 2x2 pixels. Custom post-processing refines the smallest detections before counting.
- Fine-tuned a **SAM Vision Transformer** to segment macroalgae from offshore images at 90% IoU, and trained a linear regression model that predicts plant weight within 1 gram of error.
- Generated **3D reconstructions of individual kelp blades** from multiview imagery using Structure from Motion (COLMAP, Metashape, Open3D). ArUco marker scaling recovered absolute metric volume (cm^3) and oriented bounding box dimensions, validated against hand measurements. *[view repo] [view abstract]*

Computer Vision Engineer & Co-founder, Breathily (funded by UCSF) — *March 2020 – March 2022*

- **2nd place, UC Launch Accelerator. UCSF Catalyst Program funding. NSF I-Corps participant.** Co-founded a startup developing contactless pulmonary function testing for patients with neuromuscular diseases such as ALS. The system eliminates the forced mouth seal required by standard spirometry. *[view demo]*
- Built the **computer vision pipeline** that converts chest wall motion captured by Intel RealSense depth cameras into pulmonary function values (FVC, FEV1, FEV1/FVC, PEF, FEF25-75). Components: pose estimation (Cubemos and MediaPipe), RGB-D processing (OpenCV), 1D signal analysis (SciPy), 3D mesh estimation (Pixel2Mesh), and CNN plus LSTM networks (Keras). *[view repo]*
- Co-led an **IRB-approved clinical study** at the UCSF Adult Pulmonary Function Lab that captured paired depth and reference spirometer data from live patient visits. Evaluation used correlation coefficient, RMSE, and MAE against the spirometer.
- Co-inventor, **U.S. Patent Application US20240090795A1**, "Methods for Pulmonary Function Testing With Machine Learning Analysis and Systems for Same," filed February 2022. *[view patent]*

Computer Vision Scientist, UC San Francisco Hospital (UCSF) — *November 2018 – March 2020*

Fahy Lab (June 2019 – March 2020)

- Built a **3D mucus plug measurement pipeline** on chest CT. Mask R-CNN produces per plug instance segmentation. GK fuzzy clustering isolates plug tissue at boundaries where hard thresholding introduces partial volume error. Marching cubes and PCA cylinder fitting were used to extract the length and width for each plug.
- **Segmented central airways** using a combined region growing and CNN method, then mapped each plug to its airway generation. Derived a **novel airway resistance score** that classifies plugs as stubby (≤12 mm) or stringy

(>12 mm) and correlates with FEV1 decline **across 580+ patients**. Published in JCI Insight (2023). *[view publication]*

Abbasi Lab (November 2019 – June 2020)

- Decoded patient attention/engagement levels from a 64-channel EEG. Morlet wavelet spectrograms were followed by **NMF decomposition** and linear regression. Topographic scalp maps recovered distinct activation patterns for engaged and disengaged states.

Sohn Lab (November 2018 – June 2019)

- Trained a custom **3D CNN** on brain CT to predict brain age across 8 volumetric regions. The predicted minus true age gap was fed into a GLM to screen a panel of systemic conditions (hypertension, diabetes, polysubstance abuse).

Computer Vision Researcher, UC Berkeley *May 2017 – January 2019*

- **Estimated sorghum stem width from video** captured by a moving **agricultural robotic platform**. Faster R-CNN located stems in each frame; a Wiener filter, Canny edges, morphological operations, and RANSAC recovered a clean stem boundary before depth-based metric conversion. Published in the Electronic Imaging Conference 2019. *[view publication]*

FULL-STACK AI ENGINEERING PROJECTS

FOIA Fluent, Data Science for Social Good (NYCxDSSG) *March 2026 – Present*

- Built an **open source civic AI platform** end-to-end using FastAPI, Next.js 14, Supabase (Postgres with Row Level Security and OTP auth), Claude API, and Tavily, deployed on Railway and Vercel. *[repo] [site]*
- Shipped an **AI chat assistant with 11 tools** and a 4-tier response pipeline (local lookup, trusted domain search, deep research agent, fallback). The assistant auto-escalates from Claude Haiku to Sonnet and is grounded solely in tool results.
- Built a **Transparency Hub** that aggregates FOIA outcomes across **1,600+ federal agencies, 54 state jurisdictions, and 17 years of FOIA.gov annual report data**, refreshed weekly from MuckRock and FOIA.gov.

SKILLS

Computer Vision and ML: PyTorch, TensorFlow, Keras, Hugging Face, OpenCV, scikit-image, Google Vertex AI

3D Vision: COLMAP, Metashape, Open3D, Intel RealSense, Lidar

Serving and MLOps: NVIDIA Triton, TensorRT, CUDA, Docker, GKE, Cloud Run, Google Cloud Composer, Apache Airflow, Celery

Backend and Web: FastAPI, Next.js, Supabase, Postgres, Redis

Languages: Python, C++, TypeScript, MATLAB

EDUCATION

B.A. Computer Science, University of California, Berkeley

SELECTED PUBLICATIONS, PATENTS, & ABSTRACTS

[1] B. Huang, ..., **F. Heng**, ..., J. Fahy. “Mucus plugs in proximal airway generations are consequential for airflow limitation in asthma.” *JCI Insight*, December 2023 **[Computer Vision, Health]**

[2] **F. Heng**, J. Belanger, M. Horowitz, K. Brown, C. Payne, R. Masalia. “Computer vision based phenotyping approaches in the brown macroalgae, Saccharina Latissima.” *NAPPN National Conference*, St. Louis, Missouri, October 2022 **[Computer Vision, Climate]**

[3] **F. Heng**, X. Orain. “Methods for Pulmonary Function Testing With Machine Learning and Systems for Same.” *U.S. Patent Application US20240090795A1*, filed February 2022, Patent Pending **[Computer Vision, Health]**

[4] A. Sahiner, **F. Heng**, A. Balamurugan, A. Zakhor. “In Situ Width Estimation of Biofuel Plant Stems.” *Electronic Imaging Conference*, Burlingame, California, January 2019 **[Computer Vision, Robotics]**